

Karol Mastalerz

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Education

Warsaw University of Technology (WUoT) B.Sc. Mathematics and Data Analysis - Admission via Laureate status (10th Place) in WUoT National Mathematics Competition. - Curriculum: Abstract algebra, mathematical analysis, logic, programming.	Oct 2025 – Present Warsaw, Poland
High School (Math-Physics-IT Profile) Olympic-level Mathematics Programme	2021 – 2025

Technical Skills

- Languages:** C++, Python, JavaScript (ES6+), HTML/CSS.
- AI Engineering:** Gemini (2.5, 3.0, 3.1), GLM-5, LLM-integrated workflows, prompt engineering.
- Tools:** Git, NumPy.

Technical Experience & Awards

Automated News Monitoring Application - Developed a real-time web scraping and alerting system for targeted news aggregation. - Deployed in production; actively utilized by the Local Government Department (Samorządy) at the Polish Press Agency (PAP) to monitor time-sensitive publications.	2025 – Present
Browser Game Automation Application - Engineered a standalone GUI application for game automation targeting a platform with active bot-detection. - Implemented behavioral heuristics to bypass anti-cheat mechanisms. - Developed asynchronous logic for real-time game state monitoring and response.	2022 – 2025
AI-Augmented Development (Vibecoding) - Utilization of SOTA LLMs (Gemini, GLM-5) as a translation layer for user-defined architectural logic and algorithmic proofs into executable code. - Rapid prototyping of C++ and JS components through human-led design and AI-assisted implementation.	2025 – Present
National Championships in Algorithmics and Programming - Competed in team-based competitive programming and algorithmic problem-solving. - Developed and optimized C++ solutions under strict time and memory complexity constraints.	2023 – 2024
Competitive Mathematics Instructor - Instruction of elementary students for national mathematical competitions. - Deconstruction of advanced combinatorial and algebraic concepts into accessible frameworks.	2024 – Present

Core Proficiencies

- Combinatorics & Algorithmics:** Discrete mathematics and algorithmic efficiency.
- Technical Problem Solving:** Systematic deconstruction of complex logic-based challenges.
- Rapid Knowledge Acquisition:** High-velocity learning across new technical domains.